

Lecture 38 — When $Ax = b$ has no exact solution

Mon Nov 30

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Last updated: August 18, 2026.

The class outline for this day will be posted before class.

Related reading and the course-wide list of daily objectives: [Lectures](#).

Learning objectives

By the end of class, students should be able to:

1. Recognize an inconsistent overdetermined system.
2. Formulate the least-squares objective $\min_x \|Ax - b\|$.
3. Interpret the fitted vector as the closest attainable vector.
4. Interpret the residual geometrically.
5. Distinguish an exact solution from a best approximate solution.